HEISENBERG UNCERTAINTY AND TWISTED MÖBIUS DYNAMICS VI: COHOMOLOGICAL CURVATURE, PRIME–ZETA BRANCHING, AND THE POSITIVITY BARRIER

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Remark 0.1 (Version 6). The canonical pair, Weyl calculus, Robertson inequality, and general twisted algebra are standard. The arithmetic specialization $n \mapsto (\ell \log n, \kappa\Omega(n))$ produces the specific cocycle studied here. Version 6 retains the exact Version 4 curvature results and adds: a structural obstruction showing that every expectation value of a linearly Möbius-weighted state is necessarily quadratic in the Möbius coefficients; a closed formula for the second ϑ-curvature of the twisted inverse $z_\vartheta(n)$; and an exact Dirichlet-series summation of that curvature; a precise identification of the prime-zeta logarithm as the source of its branch points; a two-variable Euler-product/Hessian representation of the curvature determinant; a proof that coefficient positivity and complete monotonicity alone cannot remove the logarithmic singularities; an exact local monodromy formula with Jordan-totient weight; a correction of the distributional transpose formulas; and multiplicative superadditivity, divisor monotonicity, and power-growth laws for the curvature.

1. Introduction: the Version 6 structure

Version 1 placed a standard Heisenberg pair next to a zeta multiplier but did not couple the two structures. Version 2 corrected that defect: the exact canonical commutation relation deformed the arithmetic product itself through a nontrivial cocycle. The Version 2 audit nevertheless identified two remaining weaknesses. First, the actual uncertainty principle was not stated prominently. Second, several domain, inverse, and convergence claims required repair.

Version 6 starts from the self-adjoint position and momentum operators

$$(Q\psi)(q) = q\psi(q), \qquad (P\psi)(q) = -i\hbar\psi'(q), \tag{1.1}$$

with the canonical commutation relation

$$[Q, P] = i\hbar I \tag{1.2}$$

on the Schwartz core. For every normalized Schwartz state, the Heisenberg–Robertson uncertainty inequality is [1]

$$\boxed{\Delta_\psi Q \, \Delta_\psi P \geq \frac{\hbar}{2}.} \tag{1.3}$$

Here the uncertainty is a mathematical spread intrinsic to the state, not an experimental error.

The arithmetic specialization assigns to an integer n the phase-space increments

$$x_n = \ell \log n, \qquad y_n = \kappa\Omega(n), \tag{1.4}$$

and the self-adjoint generator

$$G_n = y_n Q - x_n P. \tag{1.5}$$

The arithmetic specialization of the standard Robertson inequality gives, for two integers,

$$\boxed{\Delta_\psi G_m \, \Delta_\psi G_n \geq \frac{\hbar\ell\kappa}{2} \left|\Omega(m)\log n - \log m\, \Omega(n)\right|.} \tag{1.6}$$

For distinct primes p, q, this becomes

$$\boxed{\Delta_\psi G_p \, \Delta_\psi G_q \geq \frac{\hbar\ell\kappa}{2} \left|\log \frac{q}{p}\right|.} \tag{1.7}$$

Thus the same arithmetic symplectic area that later appears in the twisted Möbius product already appears as a lower bound forced directly by the Heisenberg uncertainty principle.

The associated Weyl operators are

$$W_n = \exp\left(\frac{i}{\hbar} G_n\right), \tag{1.8}$$

and satisfy

$$W_m W_n = \chi_\vartheta(m,n) W_{mn}, \tag{1.9}$$

where

$$\chi_\vartheta(m,n) = \exp\left[\frac{i\vartheta}{2}\big(\Omega(m)\log n - \log m\,\Omega(n)\big)\right], \qquad \vartheta = \frac{\ell\kappa}{\hbar}. \tag{1.10}$$

The cocycle is proved below to be cohomologically nontrivial for every $\vartheta \neq 0$. It induces an associative noncommutative Dirichlet product. Möbius inversion in that product defines coefficients $z_\vartheta(n)$. For prime powers and distinct prime pairs one obtains

$$z_\vartheta(p^r) = 1, \tag{1.11}$$

$$z_\vartheta(pq) = 2\cos\left(\frac{\vartheta}{2}\log\frac{q}{p}\right) - 1. \tag{1.12}$$

The same determinant-like arithmetic quantity therefore controls both the uncertainty lower bound (1.6) and the phase responsible for the deformed inverse.

Version 5 also differentiates the twisted inverse itself at the commuting point. For

$$n = \prod_{j=1}^{r} p_j^{a_j}, \qquad T(a) = \frac{a(a+1)}{2}, \tag{1.13}$$

one obtains

$$\boxed{-z_0''(n) = \frac{1}{2}\sum_{1\le i<j\le r} T(a_i)T(a_j)\log^2\frac{p_j}{p_i}.} \tag{1.14}$$

For square-free $n = p_1\cdots p_r$,

$$\boxed{-z_0''(n) = \frac{1}{2}\left[r\sum_{p\mid n}(\log p)^2 - (\log n)^2\right] = \frac{r^2}{2}\operatorname{Var}_{p\mid n}(\log p).} \tag{1.15}$$

Summing this curvature does not return a two-variable Möbius correlation. For $\Re s > 1$,

$$\mathcal{K}(s) = -\sum_{n\ge 1}\frac{z_0''(n)}{n^s} = \frac{\zeta(s)}{2}\big(S_0(s)S_2(s) - S_1(s)^2\big), \tag{1.16}$$

where

$$S_j(s) = \sum_p \frac{p^{-s}}{(1-p^{-s})^2}(\log p)^j. \tag{1.17}$$

Section 11 rewrites these functions through $\log\zeta(rs)$ and $\zeta'(rs)/\zeta(rs)$, producing a local continuation to $\Re s > 1/2$ and an exact singularity criterion for nontrivial zeta zeros to the right of the critical line.

Finally, a bounded self-adjoint Möbius Hamiltonian is defined for $\sigma > 1$. Its Schrödinger evolution and the exact Heisenberg equations for $Q(t)$ and $P(t)$ are derived with domains stated explicitly. The paper does not claim a connection to RH; analytic continuation, functional equations, and spectral information remain open.

Remark 1.1 (Scope of Version 6). The operator deformation is developed in the half-planes stated for each operator. Version 6 additionally develops a direct local relation between the second deformation curvature and the classical zero set in Section 12.

2. Exact canonical kinematics

2.1. Hilbert space, position, and momentum.

Definition 2.1 (Hilbert space)**.** Fix $\hbar > 0$ and set

$$\mathcal{H} = L^2(\mathbb{R}, dq) \tag{2.1}$$

with its standard complex Hilbert-space inner product.

Definition 2.2 (Position)**.** The position operator Q is

$$(Q\psi)(q) = q\psi(q) \tag{2.2}$$

with maximal domain

$$\mathcal{D}(Q) = \{\psi \in L^2(\mathbb{R}) : q\psi(q) \in L^2(\mathbb{R})\}. \tag{2.3}$$

Definition 2.3 (Momentum)**.** The momentum operator P is

$$(P\psi)(q) = -i\hbar\,\psi'(q) \tag{2.4}$$

with domain

$$\mathcal{D}(P) = H^1(\mathbb{R}). \tag{2.5}$$

Proposition 2.4 (Self-adjointness and common core)**.** *The operators Q and P are self-adjoint. The Schwartz space $\mathcal{S}(\mathbb{R})$ is a common invariant core for both.*

Proof. Q is the maximal multiplication operator by the real function q. Under the unitary Fourier transform with convention

$$(\mathcal{F}\psi)(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{\mathbb{R}} e^{-ipq/\hbar}\psi(q)\,dq, \tag{2.6}$$

P is unitarily equivalent to multiplication by the real variable p. The core statement follows by standard cutoff and mollification arguments. See, for example, [7]. □

Proposition 2.5 (Exact canonical commutation relation)**.** *For every $\psi \in \mathcal{S}(\mathbb{R})$,*

$$[Q, P]\psi = i\hbar\psi. \tag{2.7}$$

Proof. Direct differentiation gives

$$(QP\psi)(q) = -i\hbar q\psi'(q), \tag{2.8}$$

$$(PQ\psi)(q) = -i\hbar\psi(q) - i\hbar q\psi'(q). \tag{2.9}$$

Subtracting (2.9) from (2.8) proves (2.7). □

2.2. The Heisenberg uncertainty principle and exact incompatibility.

Definition 2.6 (Expectation and uncertainty)**.** Let A be self-adjoint and let $\psi \in \mathcal{D}(A)$ be normalized. Define

$$\langle A\rangle_\psi = \langle \psi, A\psi\rangle, \tag{2.10}$$

$$(\Delta_\psi A)^2 = \|(A - \langle A\rangle_\psi)\psi\|^2. \tag{2.11}$$

Theorem 2.7 (Heisenberg uncertainty principle)**.** *For every normalized $\psi \in \mathcal{S}(\mathbb{R})$,*

$$\boxed{\Delta_\psi Q\,\Delta_\psi P \geq \frac{\hbar}{2}.} \tag{2.12}$$

Proof. Set

$$\widetilde{Q} = Q - \langle Q\rangle_\psi I, \qquad \widetilde{P} = P - \langle P\rangle_\psi I. \tag{2.13}$$

By Cauchy–Schwarz,

$$\Delta_\psi Q\, \Delta_\psi P \geq \left|\langle \widetilde{Q}\psi, \widetilde{P}\psi\rangle\right|. \tag{2.14}$$

The modulus dominates the absolute value of the imaginary part, and on $\mathcal{S}(\mathbb{R})$

$$2i\ \mathrm{Im}\langle \widetilde{Q}\psi, \widetilde{P}\psi\rangle = \langle \psi, [Q,P]\psi\rangle \tag{2.15}$$

$$= i\hbar. \tag{2.16}$$

Combining (2.14)–(2.16) proves (2.12). □

Remark 2.8 (Meaning of the inequality). Equation (2.12) is the Heisenberg uncertainty principle used in this paper. The quantities $\Delta_\psi Q$ and $\Delta_\psi P$ are intrinsic standard deviations of a state. They are not measurement errors. Exact simultaneous localization would require both to vanish, which is prohibited by (2.12).

Theorem 2.9 (No simultaneous exact position and momentum)**.** *There is no nonzero state with simultaneous exact position and exact momentum. More precisely, both of the following are impossible for real q_0, p_0:*

(1) a nonzero vector ψ in a common invariant domain satisfying

$$Q\psi = q_0\psi, \qquad P\psi = p_0\psi; \tag{2.17}$$

(2) a nonzero tempered distribution $T \in \mathcal{S}'(\mathbb{R})$ satisfying

$$qT = q_0 T, \qquad -i\hbar T' = p_0 T. \tag{2.18}$$

Proof. For vectors, (2.17) gives $[Q,P]\psi = 0$, whereas the CCR gives $[Q,P]\psi = i\hbar\psi$; hence $\psi = 0$. For distributions define directly

$$QT = qT, \qquad PT = -i\hbar T'. \tag{2.19}$$

The distributional Leibniz rule gives

$$[Q,P]T = i\hbar T. \tag{2.20}$$

The simultaneous eigenvalue equations would make the left-hand side zero, so $T = 0$. □

3. Weyl displacements and the arithmetic phase space

3.1. Weyl operators.

Definition 3.1 (Weyl displacement)**.** For $(x,y) \in \mathbb{R}^2$, define

$$W(x,y) = \exp\!\left(\frac{i}{\hbar}(yQ - xP)\right). \tag{3.1}$$

This is a unitary operator on all of $\mathcal{H}$.

Proposition 3.2 (Coordinate action)**.** *For $\psi \in L^2(\mathbb{R})$,*

$$(W(x,y)\psi)(q) = \exp\!\left(\frac{i}{\hbar} y\left(q - \frac{x}{2}\right)\right)\psi(q-x) \tag{3.2}$$

almost everywhere.

Proof. Since $[iyQ/\hbar, -ixP/\hbar]$ is scalar, the Baker–Campbell–Hausdorff formula gives

$$W(x,y) = e^{-ixy/(2\hbar)} e^{iyQ/\hbar} e^{-ixP/\hbar}. \tag{3.3}$$

The last factor translates by x and the middle factor multiplies by $e^{iyq/\hbar}$, yielding (3.2). □

Proposition 3.3 (Weyl form of the CCR)**.** *Let $z = (x, y)$ and $z' = (x', y')$. Define*

$$\omega(z, z') = yx' - xy'. \tag{3.4}$$

Then

$$W(z)W(z') = \exp\left(\frac{i}{2\hbar}\omega(z, z')\right) W(z + z'). \tag{3.5}$$

Consequently,

$$[W(z), W(z')] = 2i \sin\left(\frac{\omega(z, z')}{2\hbar}\right) W(z + z'). \tag{3.6}$$

Proof. For

$$A_z = \frac{i}{\hbar}(yQ - xP), \tag{3.7}$$

the CCR gives

$$[A_z, A_{z'}] = \frac{i}{\hbar}\omega(z, z')I. \tag{3.8}$$

This commutator is central, so the Baker–Campbell–Hausdorff series stops after the commutator term and gives (3.5). Reversing z and z' changes the sign of ω, giving (3.6). The Weyl relations are standard; see [6]. □

Proposition 3.4 (Position and momentum displacement equations)**.** *On $\mathcal{S}(\mathbb{R})$,*

$$[Q, W(x, y)] = xW(x, y), \tag{3.9}$$

$$[P, W(x, y)] = yW(x, y). \tag{3.10}$$

Proof. Both identities follow by differentiating (3.2), or directly from (2.7) because the relevant commutators with the exponent are scalar. □

3.2. **Arithmetic embedding.**

Definition 3.5 (The additive prime-factor counter)**.** For

$$n = \prod_p p^{v_p(n)}, \tag{3.11}$$

define

$$\Omega(n) = \sum_p v_p(n). \tag{3.12}$$

Then

$$\Omega(mn) = \Omega(m) + \Omega(n). \tag{3.13}$$

Definition 3.6 (Arithmetic phase-space map)**.** Fix two positive scales $\ell > 0$ and $\kappa > 0$. For $n \in \mathbb{N}$, put

$$x_n = \ell \log n, \tag{3.14}$$

$$y_n = \kappa\Omega(n), \tag{3.15}$$

$$z_n = (x_n, y_n). \tag{3.16}$$

Define the dimensionless Heisenberg area parameter

$$\vartheta = \frac{\ell\kappa}{\hbar}. \tag{3.17}$$

Remark 3.7 (Why these two coordinates). The first coordinate is forced by multiplicative size because $\log(mn) = \log m + \log n$. The second coordinate is chosen as the simplest integer-valued completely additive function that directly controls the Möbius sign on square-free integers:

$$\mu(n) \neq 0 \quad \Longrightarrow \quad \mu(n) = (-1)^{\Omega(n)}. \tag{3.18}$$

The construction can be generalized by replacing Ω with another completely additive arithmetic function, but the present notebook fixes Ω throughout.

Definition 3.8 (Arithmetic Weyl operators)**.** For $n \in \mathbb{N}$, define

$$W_n = W(x_n, y_n). \tag{3.19}$$

Theorem 3.9 (Exact Heisenberg cocycle on the integers)**.** *For $m, n \in \mathbb{N}$,*

$$W_m W_n = \chi_\vartheta(m,n) W_{mn}, \tag{3.20}$$

where

$$\chi_\vartheta(m,n) = \exp\left[\frac{i\vartheta}{2}\big(\Omega(m)\log n - \log m\,\Omega(n)\big)\right]. \tag{3.21}$$

Moreover,

$$[W_m, W_n] = 2i\sin\left[\frac{\vartheta}{2}\big(\Omega(m)\log n - \log m\,\Omega(n)\big)\right] W_{mn}. \tag{3.22}$$

Proof. Complete additivity gives

$$z_{mn} = z_m + z_n. \tag{3.23}$$

Substitution of (3.14)–(3.17) into (3.5) gives (3.20); (3.22) follows from (3.6). □

Corollary 3.10 (Prime commutator)**.** *For distinct primes p, q,*

$$[W_p, W_q] = 2i\sin\left(\frac{\vartheta}{2}\log\frac{q}{p}\right) W_{pq}. \tag{3.24}$$

Thus the prime operators commute only at the resonant values

$$\vartheta \log\frac{q}{p} \in 2\pi\mathbb{Z}. \tag{3.25}$$

Remark 3.11 (Where Heisenberg enters nontrivially). Equation (3.20) is the first point where Version 2 ceases to be ordinary Dirichlet algebra in operator notation. If the canonical pair were commuting, the Weyl cocycle would be 1 and the arithmetic product below would reduce to ordinary Dirichlet convolution. For nonzero ϑ, the CCR creates order-dependent phases between different arithmetic factors.

Theorem 3.12 (The arithmetic cocycle is not gauge-trivial)**.** *Extend* log *and* Ω *additively to* $\mathbb{Q}_{>0}^{\times}$*. For every real $\vartheta \neq 0$, the cocycle χ_ϑ is not a scalar coboundary on $\mathbb{N}^{\times}$, and its extension is not a scalar coboundary on $\mathbb{Q}_{>0}^{\times}$.*

Proof. On a commutative semigroup, every scalar coboundary

$$(\delta a)(m,n) = a(m)a(n)a(mn)^{-1} \tag{3.26}$$

is symmetric in m, n. Hence a necessary condition for χ_ϑ to be a coboundary is

$$\frac{\chi_\vartheta(m,n)}{\chi_\vartheta(n,m)} = 1 \tag{3.27}$$

for all m, n. For distinct primes,

$$\frac{\chi_\vartheta(p,q)}{\chi_\vartheta(q,p)} = \exp\left(i\vartheta\log\frac{q}{p}\right). \tag{3.28}$$

If this were one simultaneously for $(p,q)=(2,3)$ and $(2,5)$, then there would be nonzero integers k,l such that

$$\vartheta\log(3/2)=2\pi k,\qquad \vartheta\log(5/2)=2\pi l. \tag{3.29}$$

Thus $\log(3/2)/\log(5/2)=k/l\in\mathbb{Q}$. Exponentiating a reduced rational relation would produce an equality between incompatible prime factorizations, contradicting unique factorization. Therefore (3.27) cannot hold for all primes. □

4. Structural filter: linear Möbius states cannot escape second order

Let $v_n\in\mathcal{H}$ be arbitrary vectors and define

$$\Psi_x=\sum_{n\le x}\mu(n)v_n. \tag{4.1}$$

Theorem 4.1 (Quadratic obstruction for all expectation-value routes)**.** *For every linear operator A for which the expression is defined,*

$$\langle\Psi_x,A\Psi_x\rangle=\sum_{m,n\le x}\mu(m)\mu(n)\langle v_m,Av_n\rangle. \tag{4.2}$$

Hence every norm, expectation value, covariance, variance, or Robertson bound obtained from a state constructed linearly with coefficients $\mu(n)$ is a quadratic Möbius form.

Proof. Expand the inner product sesquilinearly and use that $\mu(n)$ is real. □

Corollary 4.2 (State-superposition route closed)**.** *Changing the reference vector, the vectors v_n, or the observable A does not alter the quadratic dependence in* (4.2)*. A non-quadratic consequence must therefore be extracted from the algebraic deformation itself.*

5. Arithmetic uncertainty generated by the CCR

Definition 5.1 (Integer generator)**.** For $n\ge 1$, define on $\mathcal{S}(\mathbb{R})$ the symmetric operator

$$G_n^{(0)}=y_nQ-x_nP=\kappa\Omega(n)Q-\ell\log n\,P. \tag{5.1}$$

It is essentially self-adjoint on $\mathcal{S}(\mathbb{R})$; denote its self-adjoint closure by G_n. The corresponding unitary displacement is

$$W_n=\exp(iG_n/\hbar). \tag{5.2}$$

Proposition 5.2 (Arithmetic commutator)**.** *For all $m,n\ge 1$, the restrictions of G_m and G_n to $\mathcal{S}(\mathbb{R})$ satisfy*

$$[G_m,G_n]=i\hbar(x_my_n-y_mx_n)I \tag{5.3}$$

$$=i\hbar\ell\kappa\big(\log m\,\Omega(n)-\Omega(m)\log n\big)I. \tag{5.4}$$

Proof. Expand the commutator and use $[Q,P]=i\hbar I$. Terms involving $[Q,Q]$ and $[P,P]$ vanish. □

Theorem 5.3 (Arithmetic Heisenberg uncertainty law)**.** *For every normalized $\psi\in\mathcal{S}(\mathbb{R})$ and all $m,n\ge 1$,*

$$\boxed{\Delta_\psi G_m\,\Delta_\psi G_n\ge\frac{\hbar\ell\kappa}{2}\left|\Omega(m)\log n-\log m\,\Omega(n)\right|.} \tag{5.5}$$

For distinct primes p,q,

$$\boxed{\Delta_\psi G_p\,\Delta_\psi G_q\ge\frac{\hbar\ell\kappa}{2}\left|\log\frac{q}{p}\right|.} \tag{5.6}$$

For two powers of the same prime the lower bound is zero.

Proof. Apply the Robertson inequality to the self-adjoint pair G_m, G_n:

$$\Delta_\psi G_m \, \Delta_\psi G_n \geq \frac{1}{2} \left| \langle \psi, [G_m, G_n]\psi \rangle \right|. \tag{5.7}$$

Substitution of (5.4) gives (5.5). If p, q are distinct primes then $\Omega(p) = \Omega(q) = 1$, which gives (5.6). If $m = p^a, n = p^b$, the determinant $\Omega(m)\log n - \log m\,\Omega(n)$ vanishes. □

Remark 5.4 (The common invariant). Define the arithmetic symplectic area

$$\mathcal{A}(m,n) = \Omega(m)\log n - \log m\,\Omega(n). \tag{5.8}$$

The uncertainty lower bound is proportional to $|\mathcal{A}(m,n)|$, while the twisted product below carries the phase $\exp(i\vartheta\mathcal{A}(m,n)/2)$. Thus uncertainty and twisted Möbius inversion are controlled by the same arithmetic invariant, although they are distinct consequences of the CCR.

5.1. **The semiclassical and arithmetic-commutative limits are distinct.** The parameter controlling the Weyl cocycle is

$$\vartheta(\hbar, \ell, \kappa) = \frac{\ell\kappa}{\hbar}. \tag{5.9}$$

The uncertainty scale and the deformation scale therefore depend on the same three parameters in different ways.

Proposition 5.5 (Opposite fixed-scale limits)**.** *Fix $\ell > 0$ and $\kappa > 0$. For every fixed pair m, n,*

$$\frac{\hbar\ell\kappa}{2}|\mathcal{A}(m,n)| \longrightarrow 0 \qquad (\hbar \to 0^+), \tag{5.10}$$

whereas

$$|\vartheta| = \frac{\ell\kappa}{\hbar} \longrightarrow \infty. \tag{5.11}$$

Consequently the ordinary semiclassical limit $\hbar \to 0^+$ with ℓ, κ fixed does not approach the untwisted Dirichlet product.

Proof. Equation (5.10) follows directly from Theorem 5.3, while (5.11) follows from (5.9). □

Proposition 5.6 (Joint semiclassical–commutative scaling)**.** *Let $\ell = \ell(\hbar) > 0$ and $\kappa = \kappa(\hbar) > 0$. Then*

$$\vartheta(\hbar) \longrightarrow 0 \qquad (\hbar \to 0^+) \tag{5.12}$$

if and only if

$$\ell(\hbar)\kappa(\hbar) = o(\hbar). \tag{5.13}$$

Under this scaling the twisted product converges coefficientwise to the ordinary Dirichlet product, and the arithmetic uncertainty lower bound obeys

$$\frac{\hbar\ell(\hbar)\kappa(\hbar)}{2}|\mathcal{A}(m,n)| = o(\hbar^2) \tag{5.14}$$

for every fixed m, n.

Proof. The equivalence of (5.12) and (5.13) is immediate from (5.9). For fixed m, n, $\chi_\vartheta(m,n) \to 1$ as $\vartheta \to 0$, so the finite divisor sum defining $f \star_\vartheta g$ converges to ordinary Dirichlet convolution coefficientwise. Multiplying $\ell(\hbar)\kappa(\hbar) = o(\hbar)$ by $\hbar/2$ gives (5.14). □

Remark 5.7 (Two independent notions of classicality). The limit $\hbar \to 0^+$ suppresses the Robertson–Heisenberg lower bound, whereas $\vartheta \to 0$ removes the arithmetic twist. They are not the same operation. With fixed ℓ, κ they move in opposite directions; simultaneous suppression requires the joint scaling (5.13).

6. The Heisenberg-twisted Dirichlet algebra

6.1. Definition and associativity.

Definition 6.1 (Heisenberg-twisted Dirichlet convolution)**.** For arithmetic functions $f, g : \mathbb{N} \to \mathbb{C}$, define

$$(f \star_\vartheta g)(k) = \sum_{d|k} f(d)g(k/d)\chi_\vartheta(d, k/d). \tag{6.1}$$

The sum is finite for every k.

Lemma 6.2 (Cocycle identity)**.** *For all $a, b, c \in \mathbb{N}$,*

$$\chi_\vartheta(a, b)\chi_\vartheta(ab, c) = \chi_\vartheta(b, c)\chi_\vartheta(a, bc). \tag{6.2}$$

Proof. The exponent in (3.21) is the alternating bilinear form obtained from the additive map $n \mapsto z_n$. Expanding both sides using $\log(ab) = \log a + \log b$ and $\Omega(ab) = \Omega(a) + \Omega(b)$ gives the same exponent. □

Proposition 6.3 (Associative noncommutative arithmetic algebra)**.** *The product $\star_\vartheta$ is associative and has identity δ_1, where*

$$\delta_1(1) = 1, \qquad \delta_1(n) = 0 \quad (n > 1). \tag{6.3}$$

For $\vartheta \neq 0$ it is generally noncommutative.

Proof. Associativity follows from Lemma 6.2 after grouping the finite divisor sums by triples $abc = k$. The identity property follows from $\chi_\vartheta(1, n) = \chi_\vartheta(n, 1) = 1$. Noncommutativity follows from Corollary 3.10; equivalently,

$$(\delta_p \star_\vartheta \delta_q)(pq) = \chi_\vartheta(p, q) \tag{6.4}$$

whereas

$$(\delta_q \star_\vartheta \delta_p)(pq) = \chi_\vartheta(q, p) = \overline{\chi_\vartheta(p, q)}. \tag{6.5}$$

□

6.2. Weighted Banach algebra and operator representation.

Definition 6.4 (Weighted arithmetic Banach space)**.** For $\sigma > 0$, set

$$\mathcal{A}_\sigma = \left\{ f : \mathbb{N} \to \mathbb{C} : \|f\|_\sigma := \sum_{n\geq 1} \frac{|f(n)|}{n^\sigma} < \infty \right\}. \tag{6.6}$$

Proposition 6.5 (Banach-algebra estimate)**.** *For $f, g \in \mathcal{A}_\sigma$,*

$$\|f \star_\vartheta g\|_\sigma \leq \|f\|_\sigma \|g\|_\sigma. \tag{6.7}$$

Hence $(\mathcal{A}_\sigma, \star_\vartheta)$ is a unital Banach algebra.

Proof. Since $|\chi_\vartheta| = 1$,

$$\|f \star_\vartheta g\|_\sigma \leq \sum_{k\geq 1} \frac{1}{k^\sigma} \sum_{d|k} |f(d)||g(k/d)| \tag{6.8}$$

$$= \sum_{d,e\geq 1} \frac{|f(d)|}{d^\sigma} \frac{|g(e)|}{e^\sigma} \tag{6.9}$$

$$= \|f\|_\sigma \|g\|_\sigma. \tag{6.10}$$

Completeness is the completeness of weighted ℓ^1. □

Definition 6.6 (Weyl–Dirichlet representation)**.** For $f \in \mathcal{A}_\sigma$, define

$$\Pi_{\sigma,\vartheta}(f) = \sum_{n\geq 1} \frac{f(n)}{n^\sigma} W_n, \tag{6.11}$$

where the series converges absolutely in operator norm.

Proposition 6.7 (Representation theorem)**.** *For $f, g \in \mathcal{A}_\sigma$,*

$$\|\Pi_{\sigma,\vartheta}(f)\| \leq \|f\|_\sigma, \tag{6.12}$$

$$\Pi_{\sigma,\vartheta}(f)\Pi_{\sigma,\vartheta}(g) = \Pi_{\sigma,\vartheta}(f \star_\vartheta g). \tag{6.13}$$

Proof. The norm bound follows from $\|W_n\| = 1$. Absolute norm convergence permits multiplication and regrouping of the double series. The coefficient of W_k is exactly (6.1) by Theorem 3.9. □

7. Möbius inversion deformed by the CCR

7.1. The Möbius operator.

Definition 7.1 (Möbius operator)**.** For $\sigma > 1$, define

$$\mathcal{M}_{\sigma,\vartheta} = \Pi_{\sigma,\vartheta}(\mu) = \sum_{n\geq 1} \frac{\mu(n)}{n^\sigma} W_n. \tag{7.1}$$

The series converges absolutely in operator norm, so $\mathcal{M}_{\sigma,\vartheta} \in \mathcal{B}(\mathcal{H})$ and

$$\mathcal{D}(\mathcal{M}_{\sigma,\vartheta}) = \mathcal{H}. \tag{7.2}$$

Remark 7.2 (Concrete coordinate action)*.* For $\psi \in L^2(\mathbb{R})$,

$$(\mathcal{M}_{\sigma,\vartheta}\psi)(q) = \sum_{n\geq 1} \frac{\mu(n)}{n^\sigma} \exp\left[\frac{i}{\hbar} y_n \left(q - \frac{x_n}{2}\right)\right] \psi(q - x_n), \tag{7.3}$$

with convergence in L^2. Thus the operator is a Möbius-weighted superposition of simultaneous translations in position and momentum; it is not a Fourier multiplier when $\kappa \neq 0$.

7.2. Formal twisted inverse.

Proposition 7.3 (Existence and uniqueness of arithmetic inverses)**.** *Let $f : \mathbb{N} \to \mathbb{C}$ satisfy $f(1) \neq 0$. Then there exists a unique arithmetic function g such that*

$$f \star_\vartheta g = g \star_\vartheta f = \delta_1. \tag{7.4}$$

Its coefficients are determined recursively from $g(1) = f(1)^{-1}$.

Proof. For $n > 1$, the equation $(f \star_\vartheta r)(n) = 0$ contains the term $f(1)r(n)$ and otherwise only values $r(d)$ with $d < n$. It therefore constructs a unique right inverse r satisfying $f \star_\vartheta r = \delta_1$. Separately, the equations $(\lambda \star_\vartheta f)(n) = 0$ construct a unique left inverse λ satisfying $\lambda \star_\vartheta f = \delta_1$. Associativity then gives

$$\lambda = \lambda \star_\vartheta (f \star_\vartheta r) = (\lambda \star_\vartheta f) \star_\vartheta r = r, \tag{7.5}$$

so the inverse is bilateral and unique. □

Definition 7.4 (Heisenberg-zeta coefficients)**.** Let $z_\vartheta : \mathbb{N} \to \mathbb{C}$ be the unique two-sided inverse of μ for $\star_\vartheta$:

$$\mu \star_\vartheta z_\vartheta = z_\vartheta \star_\vartheta \mu = \delta_1. \tag{7.6}$$

Thus

$$z_\vartheta(1) = 1, \tag{7.7}$$

and for $n > 1$,

$$z_\vartheta(n) = -\sum_{\substack{d|n \\ d>1}} \mu(d) z_\vartheta(n/d) \chi_\vartheta(d, n/d). \tag{7.8}$$

Proposition 7.5 (Commuting boundary)**.** *At $\vartheta = 0$,*

$$z_0(n) = 1 \tag{7.9}$$

for every $n \geq 1$.

Proof. When $\vartheta = 0$, $\chi_0 \equiv 1$ and $\star_\vartheta$ is ordinary Dirichlet convolution. Equation (7.6) becomes $\mu * z_0 = \delta_1$. Classical Möbius inversion gives $z_0 = \mathbf{1}$. See [10]. □

Proposition 7.6 (Ordered-factorization formula)**.** *For $n > 1$, the twisted inverse has the finite coefficient expansion*

$$\begin{aligned} z_\vartheta(n) = \sum_{r\geq 1} (-1)^r \sum_{\substack{n_1 \cdots n_r = n \\ n_j > 1}} \left(\prod_{j=1}^{r} \mu(n_j) \right) \\ \times \exp\left[\frac{i\vartheta}{2} \sum_{1 \leq i < j \leq r} \left(\Omega(n_i) \log n_j - \log n_i \, \Omega(n_j) \right) \right]. \end{aligned} \tag{7.10}$$

Only finitely many r and ordered factorizations contribute for fixed n.

Proof. Write $\mu = \delta_1 + h$ with $h(1) = 0$. Coefficientwise the formal Neumann expansion is

$$z_\vartheta = \sum_{r \geq 0} (-h)^{\star_\vartheta r}. \tag{7.11}$$

For fixed n, a product of more than $\log_2 n$ factors exceeding one cannot equal n, so the coefficient sum is finite. Iterating the bicharacter cocycle yields the pairwise phase in (7.10). □

7.3. **A bounded inverse in a rigorous half-plane.**

Definition 7.7 (Uniform Neumann threshold)**.** Let σ_* be the unique real number > 1 satisfying

$$\frac{\zeta(\sigma_*)}{\zeta(2\sigma_*)} = 2. \tag{7.12}$$

Numerically,

$$\sigma_* = 1.578017063034\ldots. \tag{7.13}$$

Theorem 7.8 (Uniform Neumann region)**.** *For every real ϑ and every $\sigma > \sigma_*$,*

$$z_\vartheta \in \mathcal{A}_\sigma. \tag{7.14}$$

Consequently

$$\mathcal{Z}_{\sigma,\vartheta} = \Pi_{\sigma,\vartheta}(z_\vartheta) = \sum_{n \geq 1} \frac{z_\vartheta(n)}{n^\sigma} W_n \tag{7.15}$$

converges absolutely in operator norm and satisfies

$$\mathcal{M}_{\sigma,\vartheta} \mathcal{Z}_{\sigma,\vartheta} = \mathcal{Z}_{\sigma,\vartheta} \mathcal{M}_{\sigma,\vartheta} = I. \tag{7.16}$$

Proof. Write $\mu = \delta_1 + h$, where $h(1) = 0$. For $\sigma > 1$,

$$\|h\|_\sigma = \sum_{n\geq 2} \frac{\mu(n)^2}{n^\sigma} \tag{7.17}$$

$$= \frac{\zeta(\sigma)}{\zeta(2\sigma)} - 1. \tag{7.18}$$

The right-hand side is continuous and strictly decreasing from $+\infty$ to 0 as σ runs from 1^+ to $+\infty$, so σ_* is unique and $\|h\|_\sigma < 1$ exactly when $\sigma > \sigma_*$. The Banach-algebra Neumann series

$$(\delta_1 + h)^{-1} = \sum_{r=0}^{\infty} (-h)^{\star_\vartheta r} \tag{7.19}$$

then converges in $\mathcal{A}_\sigma$. Uniqueness identifies it with z_ϑ, and the representation theorem gives (7.16). □

Warning 7.9 (Exact border of the proved inverse). The Möbius operator exists as a bounded operator for every $\sigma > 1$. The argument above proves a bounded inverse only for $\sigma > \sigma_*$. At $\sigma = \sigma_*$ the Neumann norm equals one and this argument is inconclusive. No claim is made here for $1 < \sigma \leq \sigma_*$.

8. The first Heisenberg arithmetic law

Theorem 8.1 (Prime rays are rigid; distinct primes interfere)**.** *Let p, q be distinct primes and let $r \geq 1$. Then*

$$z_\vartheta(p^r) = 1, \tag{8.1}$$

$$z_\vartheta(pq) = 2\cos\left(\frac{\vartheta}{2}\log\frac{q}{p}\right) - 1, \tag{8.2}$$

$$z_\vartheta(pq) - 1 = -4\sin^2\left(\frac{\vartheta}{4}\log\frac{q}{p}\right). \tag{8.3}$$

Proof. For p^r, only the divisors 1 and p contribute to the recursion because $\mu(p^j) = 0$ for $j \geq 2$. Moreover,

$$\Omega(p)\log(p^{r-1}) - \log p\,\Omega(p^{r-1}) = 0, \tag{8.4}$$

so $\chi_\vartheta(p, p^{r-1}) = 1$. Equation (7.8) gives

$$z_\vartheta(p^r) = z_\vartheta(p^{r-1}), \tag{8.5}$$

and (8.1) follows from $z_\vartheta(1) = 1$.

For pq, the four divisors $1, p, q, pq$ give

$$0 = z_\vartheta(pq) - \chi_\vartheta(p, q) - \chi_\vartheta(q, p) + 1. \tag{8.6}$$

Since

$$\chi_\vartheta(p, q) = \exp\left(\frac{i\vartheta}{2}\log\frac{q}{p}\right), \tag{8.7}$$

and $\chi_\vartheta(q, p)$ is its complex conjugate, (8.2) follows. The identity (8.3) is the elementary formula $2(\cos u - 1) = -4\sin^2(u/2)$. □

Corollary 8.2 (Loss of ordinary multiplicativity)**.** *Generically,*

$$z_\vartheta(pq) \neq z_\vartheta(p) z_\vartheta(q) = 1. \tag{8.8}$$

Thus the Heisenberg deformation preserves every one-prime ray but destroys ordinary multiplicativity through cross-prime interference.

Corollary 8.3 (Small-area expansion)**.** *For fixed distinct primes p, q,*

$$z_\vartheta(pq) = 1 - \frac{\vartheta^2}{4}\left(\log\frac{q}{p}\right)^2 + O(\vartheta^4) \qquad (\vartheta \to 0). \tag{8.9}$$

There is no linear correction.

Proposition 8.4 (Two-prime lattice recurrence)**.** *Let $p \neq q$ be primes, set*

$$Z_{a,b} = z_\vartheta(p^a q^b), \qquad \lambda = \frac{\vartheta}{2}\log\frac{q}{p}. \tag{8.10}$$

Then for $a, b \geq 1$,

$$Z_{a,b} = e^{ib\lambda} Z_{a-1,b} + e^{-ia\lambda} Z_{a,b-1} - e^{i(b-a)\lambda} Z_{a-1,b-1}, \tag{8.11}$$

with boundary values

$$Z_{a,0} = Z_{0,b} = 1. \tag{8.12}$$

In particular,

$$z_\vartheta(p^2 q) = 1 + 2\cos(2\lambda) - 2\cos\lambda. \tag{8.13}$$

Proof. Apply the right-inverse recursion (7.8) to $p^a q^b$. Since $\mu(d) \neq 0$ only when the exponents of p, q in d are at most one, only $d = p, q, pq$ contribute. Evaluation of the three cocycle factors gives (8.11); the boundary values are Theorem 8.1 on prime rays. □

Proposition 8.5 (Evenness of the full inverse)**.** *For every $n \geq 1$,*

$$z_{-\vartheta}(n) = z_\vartheta(n). \tag{8.14}$$

Proof. Changing ϑ to $-\vartheta$ reverses the cocycle:

$$(f \star_{-\vartheta} g)(n) = (g \star_\vartheta f)(n). \tag{8.15}$$

Since z_ϑ is a two-sided inverse of μ for $\star_\vartheta$, it is a two-sided inverse of μ for the opposite product $\star_{-\vartheta}$. Uniqueness of the inverse gives (8.14). □

Proposition 8.6 (Reality for real deformation)**.** *For real ϑ and every $n \geq 1$,*

$$z_\vartheta(n) \in \mathbb{R}. \tag{8.16}$$

Proof. Complex conjugation changes χ_ϑ to $\chi_{-\vartheta}$. Since μ is real, conjugating the inverse equation shows that $\overline{z_\vartheta}$ is the inverse for the $-\vartheta$ product. Proposition 8.5 therefore gives $\overline{z_\vartheta} = z_{-\vartheta} = z_\vartheta$. □

Remark 8.7 (Why this is the non-tautological step). At $\vartheta = 0$ the inverse coefficients are identically 1. At nonzero ϑ, equation (8.3) creates a new arithmetic coefficient depending on the logarithmic separation of two primes. This dependence is forced by the symplectic area in the Weyl product and disappears if the CCR is removed. It is therefore the first result in the notebook where Heisenberg noncommutativity changes the arithmetic inverse itself rather than merely constraining an unrelated state space.

9. Cohomological curvature of the twisted Möbius inverse

Set

$$\mathcal{A}(d, e) = \Omega(d)\log e - \log d\,\Omega(e), \tag{9.1}$$

so that

$$\chi_\vartheta(d, e) = \exp\left(\frac{i\vartheta}{2}\mathcal{A}(d, e)\right). \tag{9.2}$$

Proposition 9.1 (Coefficientwise entire dependence)**.** *For each fixed $n \geq 1$, the map $\vartheta \mapsto z_\vartheta(n)$ is entire. It is even, and it is real-valued on the real axis.*

Proof. Formula (7.10) is a finite sum for fixed n, each term being an exponential $e^{ic\vartheta}$. Evenness and reality follow from Propositions 8.5 and 8.6. □

Proposition 9.2 (Second derivative of the inverse)**.** *Define*

$$g(k) = \frac{1}{4}\sum_{d|k} \mu(d)\mathcal{A}(d, k/d)^2. \tag{9.3}$$

Then

$$\boxed{z_0''(n) = \sum_{k|n} g(k).} \tag{9.4}$$

Proof. Differentiate the inverse equation

$$0 = \sum_{d|n} \mu(d) z_\vartheta(n/d) \exp\Big(\frac{i\vartheta}{2}\mathcal{A}(d, n/d)\Big) \tag{9.5}$$

twice at $\vartheta = 0$. Since $z_0 \equiv 1$ and $z_0' \equiv 0$,

$$(\mu * z_0'')(n) = \frac{1}{4}\sum_{d|n} \mu(d)\mathcal{A}(d, n/d)^2 = g(n). \tag{9.6}$$

Convolving with $\mathbf{1} = \mu^{-1}$ gives (9.4). □

Lemma 9.3 (Support collapse)**.** *Let $k = \prod_{j=1}^r p_j^{a_j}$ with distinct primes p_j. Then*

$$g(k) = 0 \qquad \text{for } r = 0,\ r = 1,\ \text{or } r \geq 3. \tag{9.7}$$

For two distinct primes,

$$\boxed{g(p^\alpha q^\beta) = -\frac{\alpha\beta}{2}\log^2\frac{q}{p}.} \tag{9.8}$$

Proof. A contributing divisor is square-free and therefore corresponds to a subset $S \subseteq \{1, \dots, r\}$. Put

$$R = \Omega(k) = \sum_j a_j, \qquad L = \log k, \qquad X_S = \sum_{j\in S} \log p_j. \tag{9.9}$$

Then

$$\mathcal{A}(d, k/d) = |S|L - RX_S. \tag{9.10}$$

Its square has Boolean degree at most two, so its full alternating difference vanishes when $r \geq 3$. For $r = 1$ the area is zero. For $r = 2$, direct evaluation of the four subsets gives

$$\sum_{d|p^\alpha q^\beta} \mu(d)\mathcal{A}(d, p^\alpha q^\beta/d)^2 = -2\alpha\beta\log^2(q/p), \tag{9.11}$$

which yields (9.8). □

Theorem 9.4 (Exact cohomological curvature)**.** *If*

$$n = \prod_{j=1}^r p_j^{a_j}, \qquad T(a) = \frac{a(a+1)}{2}, \tag{9.12}$$

then

$$\boxed{z_0''(n) = -\frac{1}{2}\sum_{1\leq i<j\leq r} T(a_i)T(a_j)\log^2\frac{p_j}{p_i}.} \tag{9.13}$$

Thus $-z_0''(n) \geq 0$, with equality exactly for $n = 1$ or a prime power.

Proof. Only divisors supported on exactly two primes survive by Lemma 9.3. Summing (9.8) over $1 \le \alpha \le a_i$, $1 \le \beta \le a_j$ gives the factor $T(a_i)T(a_j)$. □

Corollary 9.5 (Square-free prime-log variance)**.** *If $n = p_1 \cdots p_r$ is square-free, then*

$$-z_0''(n) = \frac{1}{2}\sum_{i<j}(\log p_j - \log p_i)^2 \tag{9.14}$$

$$= \frac{1}{2}\left[r\sum_{p|n}(\log p)^2 - (\log n)^2\right] \tag{9.15}$$

$$= \frac{r^2}{2}\,\mathrm{Var}_{p|n}(\log p). \tag{9.16}$$

Corollary 9.6 (Infinitesimal cohomological size)**.** *For square-free $n = p_1 \cdots p_r$, let*

$$B_{ij} = \log p_j - \log p_i. \tag{9.17}$$

Then B has rank at most two. If $r \ge 2$, its two nonzero singular values are both

$$s(n) = \sqrt{r\sum_{p|n}(\log p)^2 - (\log n)^2}, \tag{9.18}$$

and

$$\boxed{-z_0''(n) = \frac{1}{2}s(n)^2.} \tag{9.19}$$

Proof. Write $B = \mathbf{1}a^T - a\mathbf{1}^T$, $a_i = \log p_i$. The two nonzero singular values of $uv^T - vu^T$ equal

$$\sqrt{\|u\|^2\|v\|^2 - (u\cdot v)^2}. \tag{9.20}$$

Apply this with $u = \mathbf{1}$, $v = a$, and then use Corollary 9.5. □

Corollary 9.7 (Exact Taylor law for each integer)**.** *For every fixed n,*

$$z_\vartheta(n) = 1 + \frac{\vartheta^2}{2}z_0''(n) + O_n(\vartheta^4), \tag{9.21}$$

and the Taylor series has infinite radius of convergence. For square-free n,

$$\boxed{z_\vartheta(n) = 1 - \frac{\vartheta^2}{4}\left[\omega(n)\sum_{p|n}(\log p)^2 - (\log n)^2\right] + O_n(\vartheta^4).} \tag{9.22}$$

10. Multiplicative geometry of the curvature

Define

$$\mathcal{C}(n) := -z_0''(n) = \frac{1}{2}\sum_{p<q} T(v_p(n))T(v_q(n))\log^2\frac{q}{p}. \tag{10.1}$$

This section studies $\mathcal{C}$ before any summation over n.

Proposition 10.1 (Prime-ray rigidity)**.**

$$\boxed{\mathcal{C}(n) = 0 \iff n = 1 \text{ or } n = p^a.} \tag{10.2}$$

Proof. Every summand in (10.1) is nonnegative. It is strictly positive exactly when two distinct primes occur with positive valuation. □

Theorem 10.2 (Multiplicative superadditivity)**.** *For all positive integers m, n,*

$$\boxed{\mathcal{C}(mn) \geq \mathcal{C}(m) + \mathcal{C}(n).} \tag{10.3}$$

If $m, n > 1$, equality holds exactly when $m = p^a$ and $n = p^b$ for the same prime p.

Proof. Put $a_p = v_p(m)$, $b_p = v_p(n)$, $A_p = T(a_p)$, $B_p = T(b_p)$. Since $T(a+b) = T(a) + T(b) + ab$, subtraction of $\mathcal{C}(m) + \mathcal{C}(n)$ from $\mathcal{C}(mn)$ gives

$$\mathcal{C}(mn) - \mathcal{C}(m) - \mathcal{C}(n) = \frac{1}{2} \sum_{p<q} \Xi_{p,q}(m,n) \log^2 \frac{q}{p}, \tag{10.4}$$

where

$$\Xi_{p,q}(m,n) = A_p B_q + B_p A_q + (A_p + B_p) a_q b_q + (A_q + B_q) a_p b_p + a_p b_p a_q b_q. \tag{10.5}$$

All terms are nonnegative. If $m, n > 1$, all cross terms vanish exactly when both supports consist of the same single prime. □

Corollary 10.3 (Monotonicity on the divisibility poset)**.** *If $m \mid n$, then*

$$\boxed{\mathcal{C}(m) \leq \mathcal{C}(n).} \tag{10.6}$$

Proposition 10.4 (Exact power law)**.** *If $n = \prod_p p^{a_p}$, then for every integer $k \geq 1$,*

$$\boxed{\mathcal{C}(n^k) = \frac{1}{8} \sum_{p<q} \left[k^4 a_p^2 a_q^2 + k^3 (a_p^2 a_q + a_p a_q^2) + k^2 a_p a_q \right] \log^2 \frac{q}{p}.} \tag{10.7}$$

Hence, if n contains at least two distinct primes,

$$\boxed{\lim_{k \to \infty} \frac{\mathcal{C}(n^k)}{k^4} = \frac{1}{8} \sum_{p<q} a_p^2 a_q^2 \log^2 \frac{q}{p} > 0.} \tag{10.8}$$

Proof. Use $T(ka) = (k^2 a^2 + ka)/2$ in (10.1) and expand. □

Proposition 10.5 (Coprime block decomposition)**.** *Assume $(m, n) = 1$. Define*

$$W(m) = \sum_{p \mid m} T(v_p(m)), \tag{10.9}$$

$$\overline{L}(m) = \frac{1}{W(m)} \sum_{p \mid m} T(v_p(m)) \log p, \tag{10.10}$$

$$V(m) = \frac{1}{W(m)} \sum_{p \mid m} T(v_p(m)) (\log p - \overline{L}(m))^2, \tag{10.11}$$

with the analogous definitions for n. Then

$$\boxed{\mathcal{C}(mn) = \mathcal{C}(m) + \mathcal{C}(n) + \frac{1}{2} W(m) W(n) \left[V(m) + V(n) + (\overline{L}(m) - \overline{L}(n))^2 \right].} \tag{10.12}$$

Proof. The cross-prime contribution is $\frac{1}{2} \sum_{p \mid m, q \mid n} A_p B_q (\log q - \log p)^2$. Expanding around the two weighted means gives (10.12). Abstractly this last algebraic identity is the standard weighted within/between variance decomposition; its use here is only to expose the multiplicative geometry of $\mathcal{C}$. □

11. Dirichlet summation of the curvature

Define

$$c(n) = -z_0''(n) \geq 0. \tag{11.1}$$

Theorem 11.1 (Exact curvature Dirichlet series)**.** *For $\Re s > 1$,*

$$\mathcal{K}(s) = \sum_{n\geq 1} \frac{c(n)}{n^s} \tag{11.2}$$

converges absolutely and

$$\boxed{\mathcal{K}(s) = \frac{\zeta(s)}{2}\left[S_0(s)S_2(s) - S_1(s)^2\right],} \tag{11.3}$$

where

$$S_j(s) = \sum_p \frac{p^{-s}}{(1-p^{-s})^2}(\log p)^j, \qquad j = 0, 1, 2. \tag{11.4}$$

Equivalently,

$$S_0S_2 - S_1^2 = \sum_{p<q} \frac{p^{-s}q^{-s}}{(1-p^{-s})^2(1-q^{-s})^2} \log^2 \frac{q}{p}. \tag{11.5}$$

Proof. Use Theorem 9.4 and

$$\sum_{a\geq 1} T(a)x^a = \frac{x}{(1-x)^3}. \tag{11.6}$$

Factoring the unrestricted Euler factors gives $\zeta(s)$, while a selected prime contributes

$$\frac{p^{-s}}{(1-p^{-s})^2}. \tag{11.7}$$

The pair sum is exactly the weighted variance determinant $S_0S_2 - S_1^2$. Absolute convergence follows for $\Re s > 1$. □

Corollary 11.2 (Prime-zeta reduction)**.** *Let $P(s) = \sum_p p^{-s}$ for $\Re s > 1$. Then*

$$S_0(s) = \sum_{m\geq 1} mP(ms), \tag{11.8}$$

$$S_1(s) = -\sum_{m\geq 1} mP'(ms), \tag{11.9}$$

$$S_2(s) = \sum_{m\geq 1} mP''(ms). \tag{11.10}$$

Using

$$P(s) = \sum_{k\geq 1} \frac{\mu(k)}{k} \log \zeta(ks), \tag{11.11}$$

gives

$$S_0(s) = \sum_{r\geq 1} \frac{J_2(r)}{r} \log \zeta(rs), \tag{11.12}$$

$$S_1(s) = -\sum_{r\geq 1} \varphi(r) \frac{\zeta'(rs)}{\zeta(rs)}, \tag{11.13}$$

$$S_2(s) = \left(\frac{\zeta'}{\zeta}\right)'(s), \tag{11.14}$$

where $J_2(r) = r^2 \prod_{p|r}(1-p^{-2})$.

Proof. Expand $x/(1-x)^2 = \sum_{m\geq 1} m x^m$, substitute (11.11), and group by $r = mk$. The needed divisor identities are

$$r \sum_{k|r} \frac{\mu(k)}{k^2} = \frac{J_2(r)}{r}, \tag{11.15}$$

$$r \sum_{k|r} \frac{\mu(k)}{k} = \varphi(r), \tag{11.16}$$

$$r \sum_{k|r} \mu(k) = \begin{cases} 1, & r = 1, \\ 0, & r > 1. \end{cases} \tag{11.17}$$

The last identity collapses S_2 to (11.14). □

Proposition 11.3 (Singularity at $s = 1$)**.** *As* $s \to 1^+$,

$$\boxed{\mathcal{K}(s) \sim \frac{1}{2(s-1)^3} \log \frac{1}{s-1}.} \tag{11.18}$$

Proof. From (11.11),

$$P(s) = \log \frac{1}{s-1} + O(1). \tag{11.19}$$

Hence

$$S_0(s) = \log \frac{1}{s-1} + O(1), \quad S_1(s) = \frac{1}{s-1} + O(1), \quad S_2(s) = \frac{1}{(s-1)^2} + O(1). \tag{11.20}$$

Combine with $\zeta(s) \sim (s-1)^{-1}$. □

12. Continuation of the curvature and detection of zeta zeros

Theorem 12.1 (Local continuation to $\Re s > 1/2$)**.** *Let* $U \subset \{\Re s > 1/2\}$ *be simply connected and contain no zero or pole of* $\zeta(s)$*. Choose a branch of* $\log \zeta(s)$ *on* U*. Then* (11.12)–(11.14)*, substituted into* (11.3)*, continue* $\mathcal{K}$ *holomorphically to* U*.*

Proof. For $r \geq 2$, $\Re(rs) > 1$, so the corresponding terms are holomorphic and converge normally on compact subsets. The $r = 1$ terms are holomorphic by the hypothesis on U. □

Theorem 12.2 (Zeta zeros are logarithmic curvature singularities)**.** *Let* ρ *be a zero of multiplicity* $m \geq 1$ *with* $1/2 < \Re\rho < 1$*, and write*

$$\zeta(s) = c_\rho (s-\rho)^m (1 + O(s-\rho)), \qquad c_\rho \neq 0. \tag{12.1}$$

Then

$$\boxed{\mathcal{K}(s) = -\frac{c_\rho m^2}{2} (s-\rho)^{m-2} \log(s-\rho) + O\Big((s-\rho)^{m-2}\Big).} \tag{12.2}$$

For a simple zero,

$$\boxed{\mathcal{K}(s) = -\frac{\zeta'(\rho)}{2}\frac{\log(s-\rho)}{s-\rho} + O\Big(\frac{1}{s-\rho}\Big).} \tag{12.3}$$

Proof. For $r \geq 2$, the terms in (11.12)–(11.13) are holomorphic near ρ. With $t = s-\rho$, the singular pieces are

$$S_0 = m\log t + O(1), \qquad S_1 = -\frac{m}{t} + O(1), \qquad S_2 = -\frac{m}{t^2} + O(1). \tag{12.4}$$

Thus

$$S_0 S_2 - S_1^2 = -\frac{m^2 \log t}{t^2} + O(t^{-2}). \tag{12.5}$$

Multiply by $\zeta(s)/2$. □

Corollary 12.3 (Curvature criterion equivalent to RH)**.** *RH is equivalent to the absence of logarithmic branch singularities of $\mathcal{K}$ in*

$$\boxed{\frac{1}{2} < \Re s < 1.} \tag{12.6}$$

Proof. Theorem 12.2 detects every zero in the right half of the critical strip, while Theorem 12.1 gives holomorphy on zero-free neighbourhoods. The functional equation makes the absence of zeros in this half-strip equivalent to RH. □

13. Where the logarithmic branching comes from

The logarithmic singularity in Theorem 12.2 does not come from a meromorphic function such as ζ, ζ', or ζ''. It comes from the prime-zeta component of S_0.

Proposition 13.1 (Exact source of the logarithm)**.** *For $\Re s > 1$,*

$$S_0(s) = \sum_{m\geq 1} mP(ms), \qquad P(s) = \sum_{r\geq 1} \frac{\mu(r)}{r}\log\zeta(rs). \tag{13.1}$$

Consequently

$$S_0(s) = \sum_{k\geq 1} \frac{J_2(k)}{k}\log\zeta(ks). \tag{13.2}$$

If ρ is a zero of ζ of multiplicity m and $1/2 < \Re\rho < 1$, then in a punctured simply connected neighbourhood of ρ,

$$S_0(s) = m\log(s-\rho) + O(1). \tag{13.3}$$

Proof. The first identity follows by expanding $p^{-s}(1-p^{-s})^{-2} = \sum_{m\geq 1} mp^{-ms}$ and summing over primes. The second follows from Möbius inversion for the prime zeta function and collecting the coefficient of $\log\zeta(ks)$, exactly as in Corollary 11.2. If $1/2 < \Re\rho < 1$, every term with $k \geq 2$ is holomorphic near ρ, because $\Re(k\rho) > 1$. The $k = 1$ term is $\log\zeta(s) = m\log(s-\rho) + O(1)$. □

Remark 13.2 (Global branch structure). The classical prime-zeta continuation generated by

$$P(s) = \sum_{r\geq 1} \frac{\mu(r)}{r}\log\zeta(rs) \tag{13.4}$$

has logarithmic branch points at scaled poles and zeros, including points $1/r$ and ρ/r, and the line $\Re s = 0$ is a natural boundary for P; see Fröberg [8]. This global phenomenon does not invalidate Corollary 12.3: in the open half-strip $1/2 < \Re s < 1$, scaled points ρ/r with $r \geq 2$ cannot occur, because $\Re(\rho/r) \leq 1/2$. Hence the only logarithmic branches there are the unscaled $r = 1$ branches attached to zeros of ζ itself.

14. Local monodromy and the Jordan-totient weight

The branch structure is used here only locally. No globally reordered double sum over zeta zeros is asserted.

Proposition 14.1 (Local monodromy at an isolated scaled zero)**.** *Let ρ be a zero of ζ of multiplicity m_ρ, let $r \geq 1$, and set $s_0 = \rho/r$. Suppose a sufficiently small punctured neighbourhood of s_0 contains no other branch point contributing to* (13.2)*. Analytic continuation once positively around s_0 changes S_0 by*

$$\boxed{\Delta_{\rho/r} S_0 = 2\pi i\, m_\rho \frac{J_2(r)}{r}.} \tag{14.1}$$

Since S_1 and S_2 are meromorphic and hence single-valued, the corresponding monodromy of $\mathcal{K}$ is

$$\boxed{\Delta_{\rho/r} \mathcal{K}(s) = \pi i\, m_\rho \frac{J_2(r)}{r}\, \zeta(s) \left(\frac{\zeta'}{\zeta}\right)'(s).} \tag{14.2}$$

If several scaled singularities collide at the same point, their local monodromy contributions are added before applying this formula.

Proof. In (13.2), the branch of $\log \zeta(rs)$ changes by $2\pi i m_\rho$ around $s_0 = \rho/r$. Multiplication by its coefficient $J_2(r)/r$ gives (14.1). The functions S_1 and S_2 are built from logarithmic derivatives of ζ, so they are meromorphic and carry no logarithmic monodromy. Substitution in $\mathcal{K} = \zeta(S_0 S_2 - S_1^2)/2$ gives (14.2). □

Proposition 14.2 (Growth of the local branch weight)**.** *For every $r \geq 1$,*

$$\frac{r}{\zeta(2)} \leq \frac{J_2(r)}{r} = r \prod_{p \mid r} \left(1 - \frac{1}{p^2}\right) \leq r. \tag{14.3}$$

Consequently the individual local weight is of linear size in r. In particular, the unregularized cumulative weight cannot be finite; more precisely,

$$\sum_{r \leq R} \frac{J_2(r)}{r} = \frac{R^2}{2\zeta(3)} + O(R \log R). \tag{14.4}$$

Proof. The product in (14.3) is bounded below by $\prod_p (1 - p^{-2}) = 1/\zeta(2)$. For the average, use $J_2(r)/r = r \sum_{d \mid r} \mu(d) d^{-2}$, interchange the finite sums, and sum m up to R/d. The main term is $R^2 \sum_{d \geq 1} \mu(d) d^{-3}/2 = R^2/(2\zeta(3))$; elementary estimation of the remaining terms gives the displayed error. □

Remark 14.3 (Why S_0 nevertheless converges away from its branches). For fixed s with $\Re s > 0$, once $r \Re s > 1$, $\log \zeta(rs) = O(2^{-r \Re s})$. This exponential decay dominates the linear factor $J_2(r)/r$. Thus the growth of the local monodromy weight is compatible with convergence of (13.2) away from its singular points.

Warning 14.4 (No global zero-kernel identity is claimed). A formal substitution of a Hadamard expansion for $\log \zeta$ into (13.2) produces nested sums over r and zeta zeros. Those sums require prescribed symmetric ordering in the zero variable and separate convergence control in r. Version 6 does not interchange them and does not claim a globally convergent identity of the form $\sum_{\rho,\rho'} \mathcal{K}_s(\rho, \rho')$. The local monodromy Proposition 14.1 does not require such a rearrangement.

15. A two-variable Euler product and the covariance determinant

The determinant in (11.3) has an intrinsic interpretation that does not involve a linearly Möbius-weighted quantum state. For $\Re s > 1$ and u in a neighbourhood of zero define

$$\mathfrak{Z}(s,u) = \prod_p \left(1 - e^u p^{-s}\right)^{-1}, \qquad F(s,u) = \log \mathfrak{Z}(s,u). \tag{15.1}$$

Then $\mathfrak{Z}(s,0) = \zeta(s)$.

Theorem 15.1 (Hessian form of the curvature)**.** *For $\Re s > 1$,*

$$F_{uu}(s,0) = S_0(s), \tag{15.2}$$

$$-F_{su}(s,0) = S_1(s), \tag{15.3}$$

$$F_{ss}(s,0) = S_2(s). \tag{15.4}$$

Consequently

$$\boxed{\frac{2\mathcal{K}(s)}{\zeta(s)} = \det\begin{pmatrix} F_{uu}(s,0) & F_{us}(s,0) \\ F_{su}(s,0) & F_{ss}(s,0) \end{pmatrix}.} \tag{15.5}$$

Proof. Differentiate

$$F(s,u) = -\sum_p \log(1 - e^u p^{-s}) \tag{15.6}$$

termwise in a neighbourhood of $u = 0$, justified by normal convergence for $\Re s > 1$. The three derivatives are exactly (11.4). Equation (15.5) is then Theorem 11.1. □

Theorem 15.2 (Zeta-distribution covariance identity)**.** *Let $\sigma > 1$ be real and let N_σ be the zeta-distributed integer*

$$\mathbb{P}_\sigma(N_\sigma = n) = \frac{n^{-\sigma}}{\zeta(\sigma)}. \tag{15.7}$$

Then

$$\mathrm{Var}_\sigma(\Omega(N_\sigma)) = S_0(\sigma), \tag{15.8}$$

$$\mathrm{Cov}_\sigma(\Omega(N_\sigma), \log N_\sigma) = S_1(\sigma), \tag{15.9}$$

$$\mathrm{Var}_\sigma(\log N_\sigma) = S_2(\sigma). \tag{15.10}$$

Therefore

$$\boxed{\frac{2\mathcal{K}(\sigma)}{\zeta(\sigma)} = \det \mathrm{Cov}_\sigma\begin{pmatrix} \Omega(N_\sigma) \\ \log N_\sigma \end{pmatrix} > 0.} \tag{15.11}$$

Proof. Under (15.7), the prime valuations $v_p(N_\sigma)$ are independent geometric random variables with

$$\mathrm{Var}(v_p) = \frac{p^{-\sigma}}{(1 - p^{-\sigma})^2}. \tag{15.12}$$

Since

$$\Omega(N_\sigma) = \sum_p v_p(N_\sigma), \qquad \log N_\sigma = \sum_p v_p(N_\sigma) \log p, \tag{15.13}$$

independence gives (15.8)–(15.10). Strict positivity follows from

$$S_0 S_2 - S_1^2 = \sum_{p<q} w_p w_q \log^2(q/p), \qquad w_p = \frac{p^{-\sigma}}{(1 - p^{-\sigma})^2} > 0. \tag{15.14}$$

□

16. The positivity route and its exact obstruction

The coefficients $c(n) = -z_0''(n)$ are nonnegative. It is therefore natural to ask whether positivity itself forces the absence of the logarithmic branches in $1/2 < \Re s < 1$. It does not.

Proposition 16.1 (Complete monotonicity on the real axis)**.** *For every real $\sigma > 1$ and every integer $r \geq 0$,*

$$\boxed{(-1)^r \mathcal{K}^{(r)}(\sigma) = \sum_{n\geq 1} \frac{c(n)(\log n)^r}{n^\sigma} \geq 0.} \tag{16.1}$$

In particular $\mathcal{K}$ is completely monotone on $(1, \infty)$.

Proof. Differentiate the absolutely convergent Dirichlet series termwise. □

Proposition 16.2 (Positivity does not remove complex logarithmic branches)**.** *Nonnegative Dirichlet coefficients, complete monotonicity on the real axis, and the strict covariance inequality* (15.11) *do not imply a branch-free continuation of $\mathcal{K}$ to $1/2 < \Re s < 1$.*

Proof. The source term $P(s) = \sum_p p^{-s}$ itself has nonnegative Dirichlet coefficients for $\Re s > 1$, yet its continuation is expressed through $\log \zeta(rs)$ and has logarithmic branch points at scaled zeros of ζ; see (13.4). More directly, the positive-coefficient function S_0 in (13.2) is exactly the factor that contributes $m \log(s - \rho)$ at a hypothetical zero ρ with $1/2 < \Re\rho < 1$. Thus real-axis positivity and complete monotonicity do not control the monodromy of the analytic continuation. □

Corollary 16.3 (What would still be needed to prove RH)**.** *To deduce Corollary 12.3 from the curvature side, one needs an additional property that controls the complex continuation of $\mathcal{K}$, not merely the signs of its Dirichlet coefficients or the positive-semidefiniteness of the real covariance matrix in* (15.11).

Remark 16.4 (Status of the attempted step). The expected coefficient-positivity step has therefore been carried out and closed: $c(n) \geq 0$ gives complete monotonicity and a positive covariance Hessian, but those properties do not forbid the logarithmic branches whose absence is equivalent to RH. The obstruction is visible inside the same formula: S_0 has positive coefficients and is simultaneously the carrier of the prime-zeta logarithmic monodromy.

17. The Heisenberg-deformed zeta series

Definition 17.1 (Heisenberg-deformed zeta series)**.** For $\Re s > \sigma_*$, define

$$\zeta_{\mathrm{H}}(s; \vartheta) = \sum_{n\geq 1} \frac{z_\vartheta(n)}{n^s}. \tag{17.1}$$

Theorem 7.8 guarantees absolute convergence for every $\Re s > \sigma_*$.

Proposition 17.2 (Classical boundary)**.** *For $\Re s > 1$,*

$$\zeta_{\mathrm{H}}(s; 0) = \zeta(s). \tag{17.2}$$

Proof. By Proposition 7.5, $z_0(n) = 1$ for all n. Substitution into (17.1) gives the ordinary Dirichlet series for $\zeta(s)$. □

Proposition 17.3 (Even Heisenberg deformation)**.** *For $\Re s > \sigma_*$,*

$$\zeta_{\mathrm{H}}(s; -\vartheta) = \zeta_{\mathrm{H}}(s; \vartheta). \tag{17.3}$$

Proof. This follows termwise from Proposition 8.5 and absolute convergence. □

Warning 17.4 (No analytic continuation is asserted). The series (17.1) is absolutely convergent and holomorphic on the open half-plane $\Re s > \sigma_*$. No meromorphic continuation, functional equation, scalar Euler product, or critical-strip theory for $\zeta_{\mathrm{H}}(s;\vartheta)$ is asserted here.

Proposition 17.5 (Conditional first-order rigidity of simple zeros)**.** *Assume that, for some simple zero ρ of the ordinary zeta function, there exists a function $F(s,\vartheta)$ analytic in a neighbourhood of $(\rho,0)$ such that*

$$F(s,0) = \zeta(s), \tag{17.4}$$

$$F(s,-\vartheta) = F(s,\vartheta), \tag{17.5}$$

and that F agrees with ζ_{H} wherever both are defined. Then the local zero branch $\rho(\vartheta)$ through ρ satisfies

$$\rho'(0) = 0. \tag{17.6}$$

Proof. The implicit-function theorem applies because $\zeta'(\rho) \neq 0$. Evenness (17.5) makes the unique local zero branch even, so its first derivative at the origin vanishes. □

Warning 17.6. Proposition 17.5 is conditional on an analytic continuation in both variables and does not supply such a continuation. It is included because it identifies a precise way in which the exact $\vartheta \mapsto -\vartheta$ symmetry would constrain the motion of classical Riemann zeros under the Heisenberg deformation if that continuation exists.

18. A genuine Möbius Hamiltonian

18.1. **Definition and domain.**

Definition 18.1 (Cosine and sine Weyl quadratures)**.** For $n \geq 1$, set

$$C_n = \frac{W_n + W_n^*}{2}, \tag{18.1}$$

$$J_n = \frac{W_n - W_n^*}{2i}. \tag{18.2}$$

Both are bounded self-adjoint operators with norm at most 1.

Definition 18.2 (Möbius Hamiltonian)**.** Fix an energy scale $E_0 > 0$ and $\sigma > 1$. Define

$$H_{\mu,\sigma,\vartheta} = E_0\,(\operatorname{Re}\mathcal{M}_{\sigma,\vartheta} - I) = E_0 \sum_{n\geq 2} \frac{\mu(n)}{n^\sigma} C_n. \tag{18.3}$$

The series converges absolutely in operator norm.

Proposition 18.3 (Self-adjoint bounded dynamics)**.** *For $\sigma > 1$, $H_{\mu,\sigma,\vartheta}$ is bounded and self-adjoint on all of $\mathcal{H}$:*

$$\mathcal{D}(H_{\mu,\sigma,\vartheta}) = \mathcal{H}. \tag{18.4}$$

It therefore generates the unitary group

$$U(t) = \exp\!\left(-\frac{i}{\hbar} t H_{\mu,\sigma,\vartheta}\right). \tag{18.5}$$

Proof. The norm series in (18.3) is dominated by $E_0 \sum_{n\geq 2} |\mu(n)| n^{-\sigma}$, which converges for $\sigma > 1$. Every C_n is self-adjoint, hence so is the norm limit. The spectral theorem then defines (18.5). □

Proposition 18.4 (Explicit Schrödinger equation)**.** *For $\psi_0 \in \mathcal{H}$, the evolution $\psi(t) = U(t)\psi_0$ satisfies*

$$i\hbar\,\partial_t \psi(t,q) = (H_{\mu,\sigma,\vartheta}\psi(t))(q) \tag{18.6}$$

in the strong L^2 sense, where

$$(H_{\mu,\sigma,\vartheta}\psi)(q) = \frac{E_0}{2}\sum_{n\geq 2}\frac{\mu(n)}{n^\sigma}\Bigg[e^{\frac{i}{\hbar}y_n(q-x_n/2)}\psi(q-x_n) + e^{-\frac{i}{\hbar}y_n(q+x_n/2)}\psi(q+x_n)\Bigg]. \tag{18.7}$$

Proof. Equation (18.7) follows from Proposition 3.2 applied to W_n and W_n^*. The series converges in operator norm, so the bounded Hamiltonian gives a strong Schrödinger evolution for every initial vector. □

18.2. **Exact equations of motion for position and momentum.**

Lemma 18.5 (Bounded first commutators)**.** *For $\sigma > 1$, the series*

$$\sum_{n\geq 2}\frac{|\mu(n)|x_n}{n^\sigma}, \tag{18.8}$$

$$\sum_{n\geq 2}\frac{|\mu(n)|y_n}{n^\sigma} \tag{18.9}$$

converge. Hence $[H_{\mu,\sigma,\vartheta}, Q]$ and $[H_{\mu,\sigma,\vartheta}, P]$, initially defined on $\mathcal{S}(\mathbb{R})$, extend to bounded operators.

Proof. The first series is bounded by a constant multiple of $\sum n^{-\sigma}\log n$. Since

$$\Omega(n) \leq \frac{\log n}{\log 2}, \tag{18.10}$$

the second is bounded by another constant multiple of the same convergent series. □

Theorem 18.6 (Exact Heisenberg equations of motion)**.** *Let*

$$Q(t) = U(t)^* Q U(t), \tag{18.11}$$

$$P(t) = U(t)^* P U(t). \tag{18.12}$$

The bounded commutators in Lemma 18.5 imply invariance of $\mathcal{D}(Q)$ and $\mathcal{D}(P)$ under $U(t)$. Indeed, for $A = Q$ or P and $\psi \in \mathcal{D}(A)$, Duhamel's formula first on a core and then by graph-norm closure gives

$$AU(t)\psi = U(t)A\psi + \frac{i}{\hbar}\int_0^t U(t-\tau)[H,A]U(\tau)\psi\, d\tau,$$

whose right-hand side lies in L^2 because $[H, A]$ is bounded. On these domains, and equivalently as quadratic-form identities, the Heisenberg equations are

$$\dot{Q}(t) = \frac{E_0}{\hbar}U(t)^*\left[\sum_{n\geq 2}\frac{\mu(n)x_n}{n^\sigma}J_n\right]U(t), \tag{18.13}$$

$$\dot{P}(t) = \frac{E_0}{\hbar}U(t)^*\left[\sum_{n\geq 2}\frac{\mu(n)y_n}{n^\sigma}J_n\right]U(t). \tag{18.14}$$

At $t = 0$ this becomes

$$\dot{Q}(0) = \frac{E_0\ell}{\hbar}\sum_{n\geq 2}\frac{\mu(n)\log n}{n^\sigma}J_n, \tag{18.15}$$

$$\dot{P}(0) = \frac{E_0\kappa}{\hbar}\sum_{n\geq 2}\frac{\mu(n)\Omega(n)}{n^\sigma}J_n. \tag{18.16}$$

Proof. For $A = Q$ or P, the bounded-commutator theorem for unitary dynamics gives

$$\dot{A}(t) = \frac{i}{\hbar} U(t)^* [H_{\mu,\sigma,\vartheta}, A] U(t). \tag{18.17}$$

From Proposition 3.4,

$$[Q, C_n] = i x_n J_n, \tag{18.18}$$

$$[P, C_n] = i y_n J_n. \tag{18.19}$$

Therefore

$$[H_{\mu,\sigma,\vartheta}, Q] = -iE_0 \sum_{n\geq 2} \frac{\mu(n) x_n}{n^\sigma} J_n, \tag{18.20}$$

$$[H_{\mu,\sigma,\vartheta}, P] = -iE_0 \sum_{n\geq 2} \frac{\mu(n) y_n}{n^\sigma} J_n. \tag{18.21}$$

Lemma 18.5 justifies the sums as bounded operators. Substitution into (18.17) gives (18.13) and (18.14). □

Remark 18.7 (Physical and arithmetic meaning of the two equations). The position variable is Q, the logarithmic-scale coordinate. The momentum variable is $P = -i\hbar\partial_q$, the generator conjugate to that coordinate. Multiplication by an integer n is represented by a phase-space displacement of size $\ell \log n$ in Q and $\kappa\Omega(n)$ in P. Equations (18.15) and (18.16) therefore say that the Möbius Hamiltonian drives position by logarithmic size and momentum by prime-factor count, with the same quantum current operators J_n.

18.3. **The CCR is visible in the dynamics.**

Proposition 18.8 (Pairwise dynamical noncommutativity)**.** *Extend* log *and* Ω *additively to positive rationals and define* W_r *by* (3.19) *for* $r \in \mathbb{Q}_{>0}$*. For* $m, n \in \mathbb{N}$*, put*

$$\Phi_{m,n} = \vartheta(\Omega(m)\log n - \log m\, \Omega(n)). \tag{18.22}$$

Then

$$[C_m, J_n] = \frac{1}{2} \sin\!\left(\frac{\Phi_{m,n}}{2}\right) \Big(W_{mn} + W_{m/n} - W_{n/m} - W_{1/(mn)}\Big). \tag{18.23}$$

Proof. Expand C_m and J_n using (18.1)–(18.2). Apply (3.6) to the four pairs (z_m, z_n), $(z_m, -z_n)$, $(-z_m, z_n)$, and $(-z_m, -z_n)$. □

Corollary 18.9 (Second-order arithmetic commutator)**.** *At* $t = 0$*,*

$$\ddot{Q}(0) = \frac{iE_0^2}{2\hbar^2} \sum_{m,n\geq 2} \frac{\mu(m)\mu(n) x_n}{(mn)^\sigma} \sin\!\left(\frac{\Phi_{m,n}}{2}\right) \times \Big(W_{mn} + W_{m/n} - W_{n/m} - W_{1/(mn)}\Big), \tag{18.24}$$

and the analogous momentum equation is

$$\ddot{P}(0) = \frac{iE_0^2}{2\hbar^2} \sum_{m,n\geq 2} \frac{\mu(m)\mu(n) y_n}{(mn)^\sigma} \sin\!\left(\frac{\Phi_{m,n}}{2}\right) \times \Big(W_{mn} + W_{m/n} - W_{n/m} - W_{1/(mn)}\Big). \tag{18.25}$$

The double sums converge in operator norm for $\sigma > 1$*.*

Proof. Differentiate (18.13) and (18.14) once more and use Proposition 18.8. Polynomial logarithmic weights are summable against $(mn)^{-\sigma}$ for $\sigma > 1$. □

Corollary 18.10 (The interaction disappears with the Weyl cocycle)**.** *If $\vartheta = 0$, then*

$$\Phi_{m,n} = 0 \tag{18.26}$$

for all m, n, and all pairwise terms in (18.24)–(18.25) *vanish. Thus the second-order cross-arithmetic second-order term displayed above is generated entirely by the exact Heisenberg cocycle.*

19. Exact non-localization inside the arithmetic dynamics

Proposition 19.1 (Schwartz action and distributional transpose)**.** *For $\sigma > 1$, the Hamiltonian $H_{\mu,\sigma,\vartheta}$ maps $\mathcal{S}(\mathbb{R})$ continuously into itself. Its continuous transpose $H^{\times} : \mathcal{S}'(\mathbb{R}) \to \mathcal{S}'(\mathbb{R})$ is defined by*

$$\langle H^{\times} T, \phi \rangle = \langle T, H_{\mu,\sigma,\vartheta} \phi \rangle, \qquad T \in \mathcal{S}'(\mathbb{R}), \quad \phi \in \mathcal{S}(\mathbb{R}). \tag{19.1}$$

Then $H^{\times}$ is continuous on $\mathcal{S}'(\mathbb{R})$.

Proof. For every Schwartz seminorm $p_{a,b}(\phi) = \sup_q |q^a \phi^{(b)}(q)|$, the coordinate action of a Weyl operator gives

$$p_{a,b}(W_n \phi) \le C_{a,b}(1 + |x_n| + |y_n|)^{N_{a,b}} \sum_{j \le b} p_{a+j,b-j}(\phi) \tag{19.2}$$

for suitable constants. Since $x_n, y_n = O(\log n)$ and

$$\sum_{n \ge 2} n^{-\sigma} (1 + \log n)^N < \infty \qquad (\sigma > 1) \tag{19.3}$$

for every fixed N, the Hamiltonian series converges in every Schwartz seminorm and defines a continuous map $\mathcal{S} \to \mathcal{S}$. Therefore (19.1) defines the standard continuous dual action on $\mathcal{S}'$. No termwise extension to all tempered distributions is assumed without this transpose construction. □

Proposition 19.2 (No exact position point is a stationary generalized mode)**.** *Let $\sigma > 1$, $\ell > 0$, and $E_0 > 0$. For no $q_0 \in \mathbb{R}$ and no scalar $E \in \mathbb{C}$ does $\delta_{q_0} \in \mathcal{S}'(\mathbb{R})$ satisfy*

$$H^{\times}_{\mu,\sigma,\vartheta} \delta_{q_0} = E \delta_{q_0}. \tag{19.4}$$

Proof. The transpose action of a Weyl displacement on a point mass is

$$W(x,y)^{\times} \delta_{q_0} = \exp\left(\frac{i}{\hbar} y (q_0 - x/2) \right) \delta_{q_0 - x}, \tag{19.5}$$

$$(W(x,y)^*)^{\times} \delta_{q_0} = \exp\left(-\frac{i}{\hbar} y (q_0 + x/2) \right) \delta_{q_0 + x}. \tag{19.6}$$

The $n = 2$ term therefore contributes nonzero masses at $q_0 \pm \ell \log 2$. No other term produces the same support point as the unique $n = 2$ contribution at either $q_0 - \ell \log 2$ or $q_0 + \ell \log 2$. Since $\mu(2) = -1$, the corresponding coefficient cannot cancel. Hence the support of the left side of (19.4) contains a point distinct from q_0, while the right side is supported only at q_0. □

Remark 19.3 (Relation to exact Heisenberg impossibility)*.* Theorem 2.9 is a universal consequence of the CCR. Proposition 19.2 is more specific: the Möbius Hamiltonian itself does not admit an exactly position-localized stationary mode. The reason is not a measurement uncertainty but the exact translation content of the arithmetic Weyl operators.

20. Literature boundary and novelty status in Version 6

The literature boundary is stated component by component rather than by assigning novelty to the entire package.

20.1. **Known framework.** The following are classical or direct specializations of classical theory:

(1) the Schrödinger pair, CCR, Weyl operators, and the Robertson inequality for arbitrary observables [1, 6];
(2) projective representations and multipliers of abelian groups, including classification through alternating commutator bicharacters [2, 3];
(3) twisted group and semigroup algebras [9, 4];
(4) classical Möbius inversion, Jordan totients, and additive functions such as Ω [10];
(5) the prime-zeta identity $P(s) = \sum_{r\geq 1} \mu(r) \log \zeta(rs)/r$, its scaled logarithmic singularities, and its natural boundary at $\Re s = 0$ [8];
(6) the weighted variance decomposition used in (10.12).

In particular, Theorem 5.3 is not a new uncertainty principle: it is Robertson's inequality applied to the specific quadratures G_m, G_n.

20.2. **Notebook-specific formulas not located in the targeted search.** The targeted literature search carried out for Version 6 did not locate the following exact formulas as previously published results:

(1) the arithmetic embedding $n \mapsto (\ell \log n, \kappa\Omega(n))$ combined with twisted Möbius inversion;
(2) the resulting coefficients $z_\vartheta(n)$, including the prime-pair and two-prime recurrences;
(3) the exact curvature formula (9.13) and its prime-log variance form;
(4) the curvature Dirichlet determinant $\mathcal{K} = \zeta(S_0 S_2 - S_1^2)/2$;
(5) the local monodromy formula (14.2) with weight $m_\rho J_2(r)/r$;
(6) the multiplicative superadditivity, prime-ray rigidity, and power laws (10.3)–(10.8).

This is a priority statement only in the limited sense "not located in the search performed"; it is not a proof of bibliographic novelty.

20.3. **What is specifically new mathematics inside this manuscript.** Independently of priority, the deductions that are new relative to the standard ingredients listed above are the exact consequences proved here for the chosen arithmetic cocycle: the twisted inverse, its coefficientwise curvature, the closed prime-pair geometry, the determinant Dirichlet series, its local branch weight, and the multiplicative laws of $\mathcal{C}(n)$. By contrast, the ambient Weyl/cohomology machinery and the abstract variance identities are not new.

21. Corrections and extensions carried into Version 6

Remark 21.1. Version 6 incorporates the following mathematical corrections and clarifications: the cocycle is proved nontrivial rather than merely described as noncommutative; no faithfulness of $\Pi_{\sigma,\vartheta}$ is claimed; left and right inverses are constructed separately before being identified; the uniform Neumann half-plane is sharpened to $\sigma > \sigma_*$; the ordered-factorization formula and a two-prime recurrence are included; real-valuedness of $z_\vartheta(n)$ for real ϑ is proved; the deformed Dirichlet series is asserted holomorphic only on its proved open half-plane; the unitary evolution is not claimed to preserve Schwartz space; and the action on tempered distributions is defined by continuous transposition; the signs and supports of the two point-mass transpose formulas are corrected explicitly; and no global rearrangement of conditionally convergent sums over zeta zeros is asserted.

22. Open problems

Question 22.1 (Analytic continuation)**.** Does $\zeta_{\mathrm{H}}(s;\vartheta)$ admit a natural meromorphic continuation from $\Re s \geq 2$ to a larger domain for fixed nonzero ϑ?

Question 22.2 (Functional symmetry)**.** Is there a completion of $\zeta_{\mathrm{H}}(s;\vartheta)$ with a symmetry that reduces at $\vartheta = 0$ to the classical reflection $s \mapsto 1-s$? A nontrivial answer would be required before using the deformation to say anything substantive about the critical line.

Question 22.3 (Spectral zeros)**.** Can the zero set of a continued ζ_{H} be realized as a spectral or resonance set of an operator built intrinsically from $H_{\mu,\sigma,\vartheta}$, rather than as zeros inserted into a multiplier?

Question 22.4 (Classical-zero rigidity)**.** If an analytic two-variable continuation through $\vartheta = 0$ exists, does the exact evenness in ϑ impose any second-order constraint on the motion of the classical Riemann zeros beyond Proposition 17.5?

Question 22.5 (Choice of the second additive coordinate)**.** Which completely additive arithmetic functions $A(n)$, replacing $\Omega(n)$ in $y_n = \kappa A(n)$, produce inequivalent deformations? The choice $A = \Omega$ is distinguished here by its direct relation to the sign of μ on square-free integers, but a classification is open.

Question 22.6 (Higher cohomological curvatures)**.** Determine closed formulas for

$$\frac{d^{2k}}{d\vartheta^{2k}} z_\vartheta(n)\Big|_{\vartheta=0}, \qquad k \geq 2, \tag{22.1}$$

and for their Dirichlet generating series. The fourth-order case is the first test of whether the collapse to finitely many one-prime moments persists or irreducible higher prime geometry appears.

23. Notebook history

Remark 23.1. Version 1 used Möbius-weighted logarithmic translations, Fourier multipliers, an eta multiplier in the critical strip, generalized plane-wave null modes, and a standard CCR non-localization statement. Its main defect was that the arithmetic multiplier and the Heisenberg pair were juxtaposed rather than coupled.

Remark 23.2. Version 2 removes the critical-strip multiplier as the central mechanism and replaces ordinary Dirichlet convolution by the Weyl-cocycle product (6.1). Heisenberg noncommutativity now changes the Möbius inverse coefficients themselves and enters the exact equations of motion of the Möbius Hamiltonian.

24. Literature status of the Version 4/5 formulas

The general ingredients are established mathematics. Twisted products by 2-cocycles and projective representations belong to the standard theory of twisted group algebras; see, for example, Packer–Raeburn [9]. The prime-zeta identity

$$P(s) = \sum_{r\geq 1} \frac{\mu(r)}{r} \log \zeta(rs) \tag{24.1}$$

and the associated logarithmic branch structure and natural boundary are classical; see Fröberg [8].

The specific chain developed here—the Heisenberg arithmetic cocycle based on $(\log n, \Omega(n))$, its twisted Möbius inverse $z_\vartheta(n)$, the closed coefficient curvature (9.13), and its Dirichlet summation (11.3)—is not identified here with a prior theorem. The classical prime-zeta formulas explain the continuation of the resulting S_j, but they do not by themselves produce the curvature formula or the Hessian identity (15.5).

Remark 24.1. Version 6 retains the closure of the linear-state route and adds the exact prime-zeta branching source, Hessian/covariance representation, and positivity obstruction and moves the analysis to the cocycle itself. Differentiating the twisted Möbius inverse at $\vartheta = 0$ produces the prime-factor curvature (9.13). Summing it yields (11.3), whose local continuation detects every zeta zero in $\Re s > 1/2$ through (12.2).

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